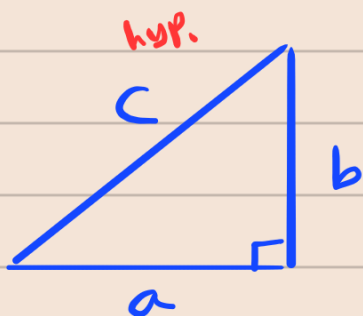


Trigonometry Recap

Right-angled triangle



$$c = \sqrt{b^2 + a^2}$$

$$a = \sqrt{c^2 - b^2}$$

$$b = \sqrt{c^2 - a^2}$$

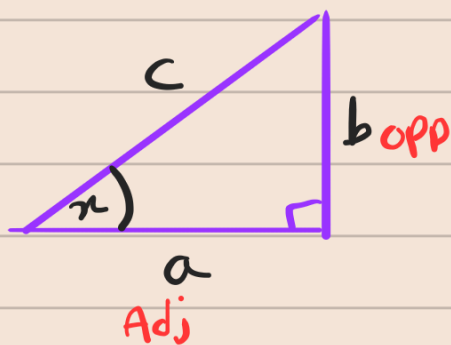
* it helps in finding a missing side while you the other 2 sides.

→ SOH CAH TO A

(1 side, 1 angle)

①

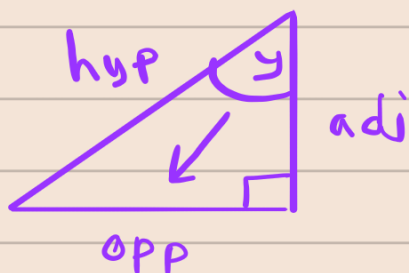
→ Missing side??



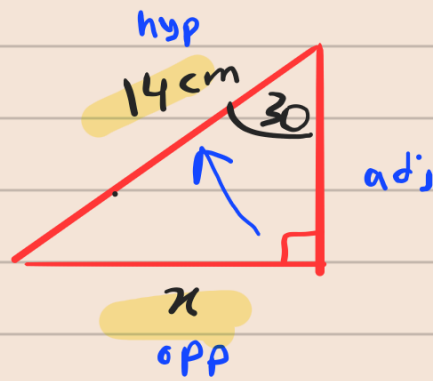
$$\sin(x) = \frac{\text{opp}}{\text{hyp}} = \frac{b}{c}$$

$$\cos(x) = \frac{\text{adj}}{\text{hyp}} = \frac{a}{c}$$

$$\tan(x) = \frac{\text{opp}}{\text{adj}} = \frac{b}{a}$$



Ex:



[3]

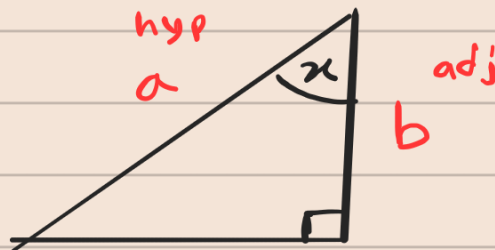
$$\sin(30) = \frac{x}{14}$$

$$x = 14 \times \sin(30) = 14 \times \frac{1}{2} = 7 \text{ cm}$$

② SOH CAHTOA (1 side, 1 side)

→ Missing angle??

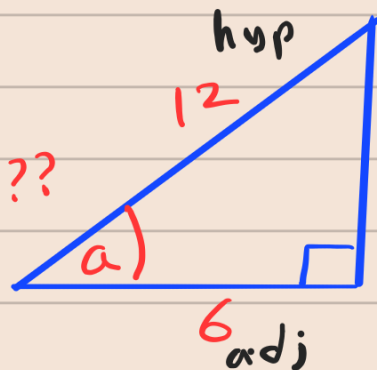
$$\cos(x) = \frac{b}{a}$$



$$x = \cos^{-1}\left(\frac{b}{a}\right)$$

Ex:

Find the size of a??

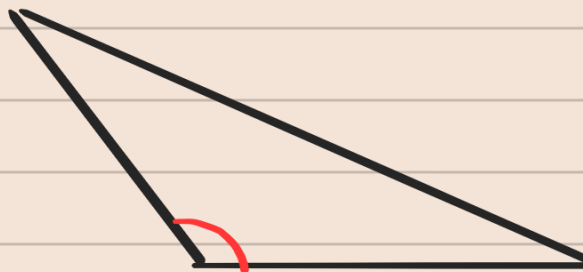


$$\cos(a) = \frac{6}{12}$$

$$\cos(a) = \frac{1}{2}$$

$$a = \cos^{-1}\left(\frac{1}{2}\right) = \boxed{60^\circ}$$

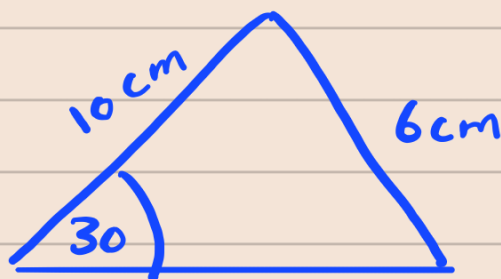
⊛ Non Right triangles



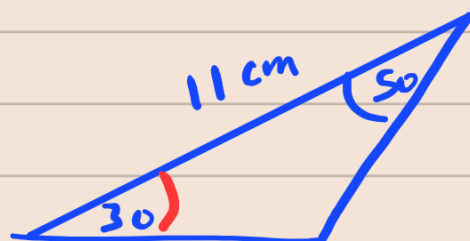
Sine law:

Cases:

- ① 2 Side lengths given, 1 not included

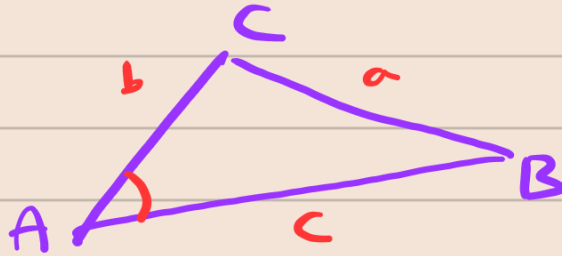


- ② 2 angles, 1 side



Formula:

$$\frac{\overset{\text{side}}{a}}{\underset{\text{angle}}{\sin A}} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

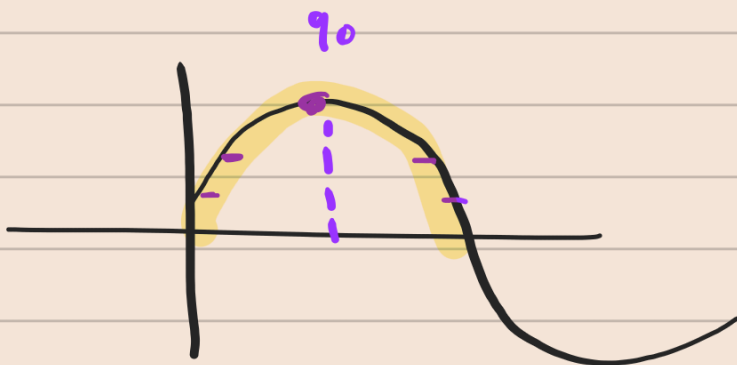


$$\frac{CB}{\sin A} = \frac{AB}{\sin C} = \frac{AC}{\sin B}$$

→ $\angle ACB$ is obtuse

Changing $\angle ACB$ [acute] to $\angle ACB$ [Obtuse]

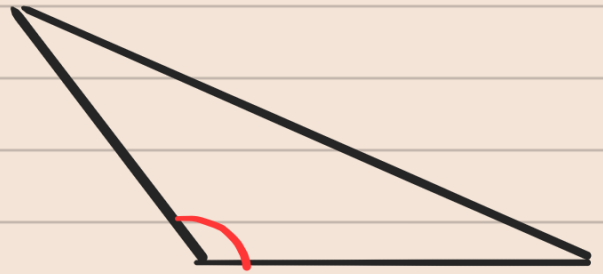
$$180 - \text{Acute} = \text{Obtuse}$$



(*) Cosine law



Acute



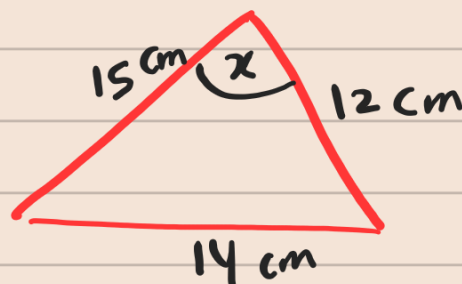
Obtuse

Cases:

- ① 2 sides given, 1 angle is included
(Missing side)



- ② 3 Sides are given (Missing angle)



→ Formula:

Sides

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$b^2 = a^2 + c^2 - 2ac \cos B$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

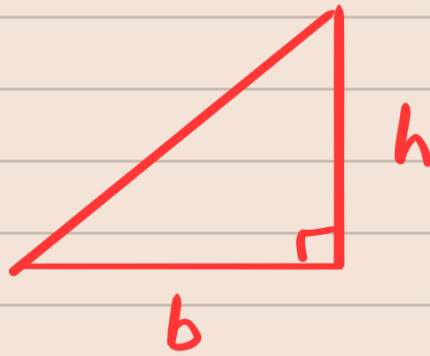
Angles

$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}$$

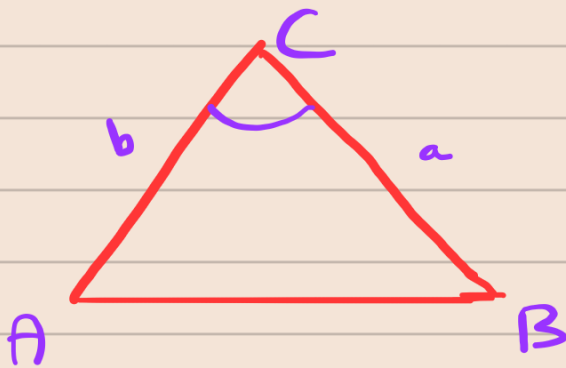
$$\cos B = \frac{a^2 + c^2 - b^2}{2ac}$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

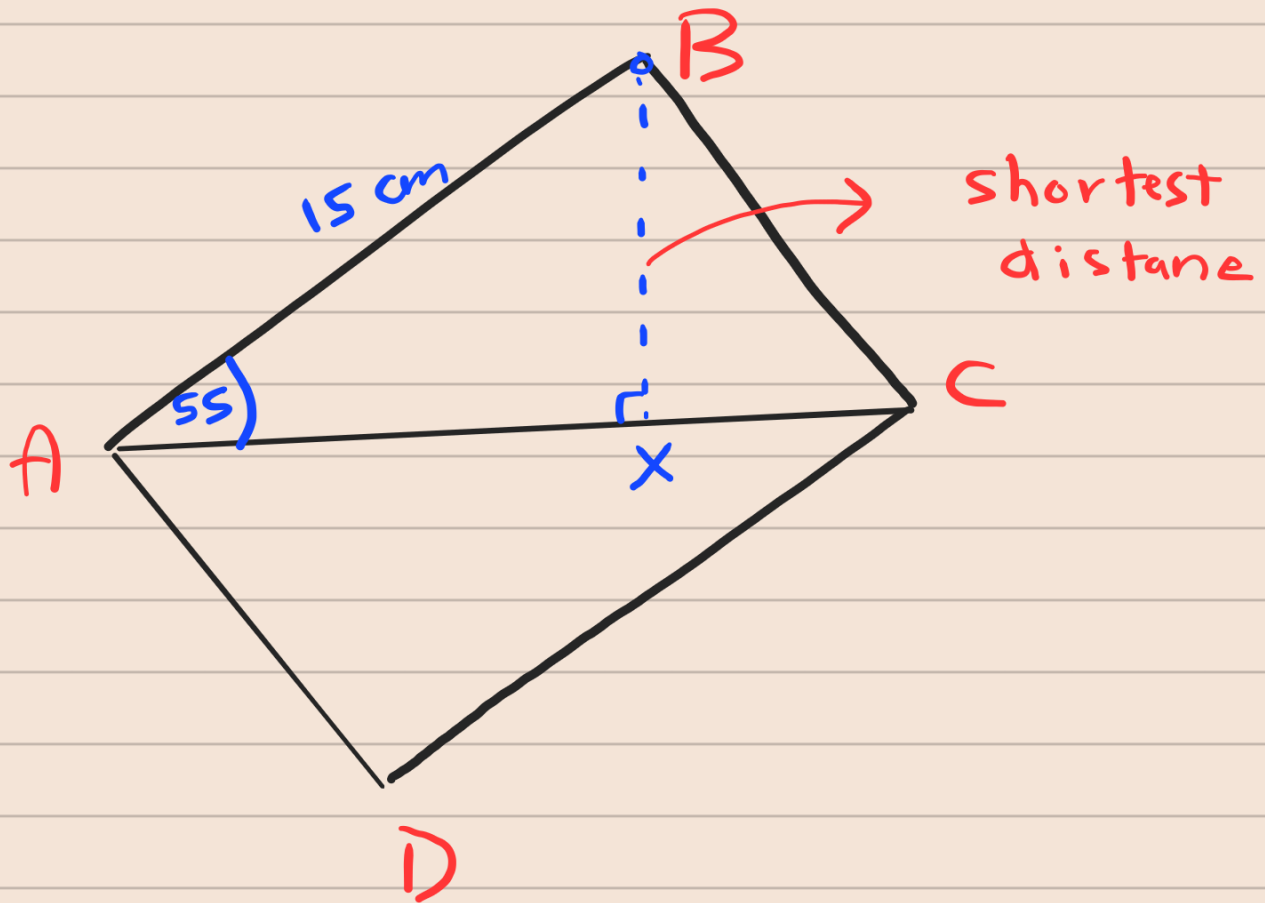
⊗ Area of triangle = $\frac{1}{2} b h$



⊗ Area of triangle = $\frac{1}{2} a b \sin C$

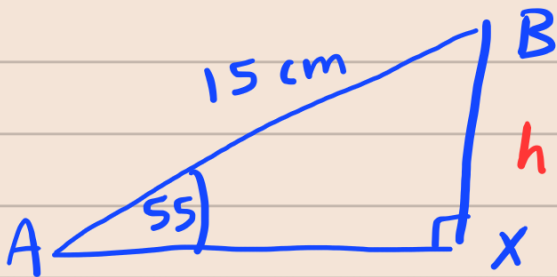


* Shortest distance



* Find the shortest distance of (B)
from AC ??

→ Draw Perpendicular from B to AC

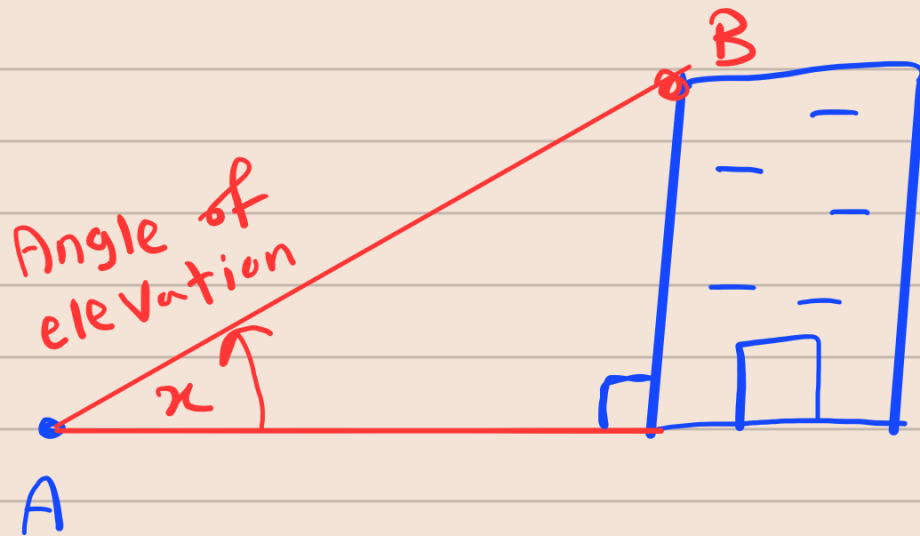


$$\sin(55) = \frac{BX}{15}$$

$$BX = 15 \times \sin(55)$$

$$= 12.28 \approx 12.3 \text{ cm}$$

* Angles of elevation & depression



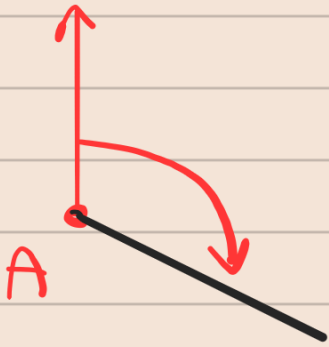
→ Angle of elevation of B from A



→ Angle of depression of A from B

* Bearings

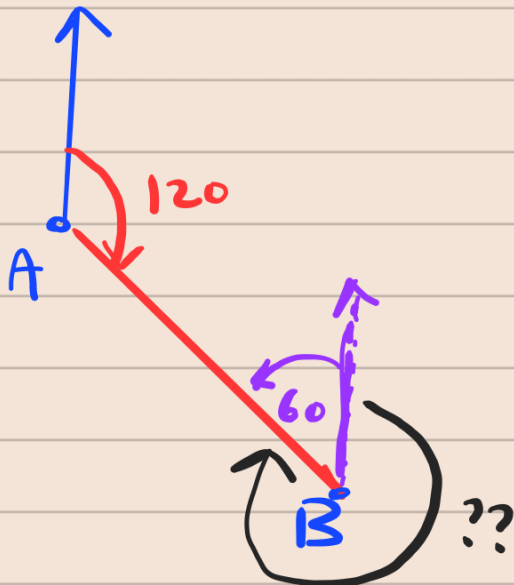
It's the angle that determines the position of a point from another point



- ① Measure From North direction
- ② Measure Clockwise
- ③ 3 digits form



• Bearing of B From A is 120°

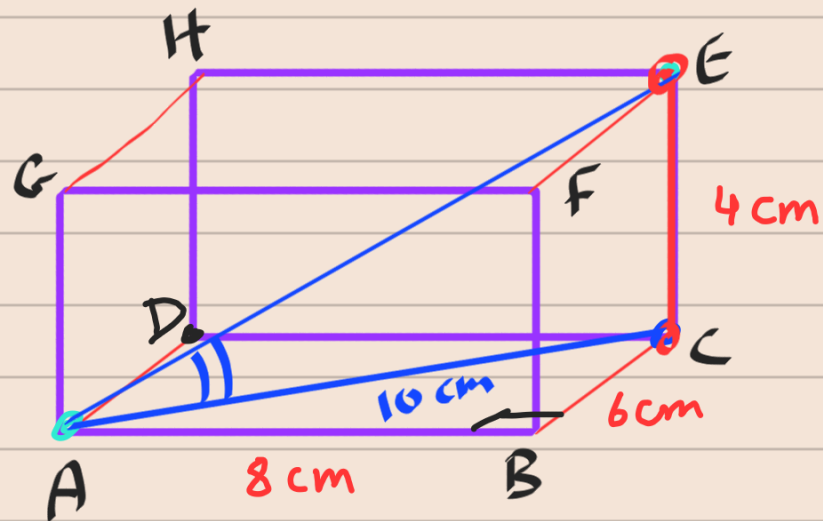


→ Bearing of A from B

$$180 - 120 = 60$$

$$360 - 60 = \boxed{300}$$

3D - Trigonometry



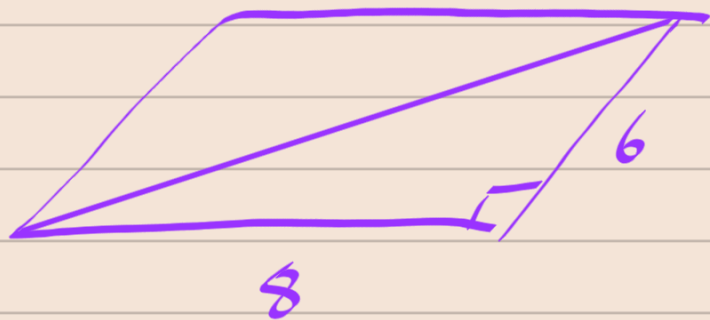
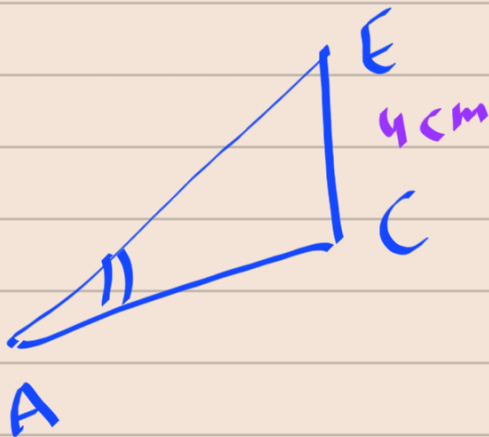
$$AB = 8 \text{ cm}$$

$$BC = 6 \text{ cm}$$

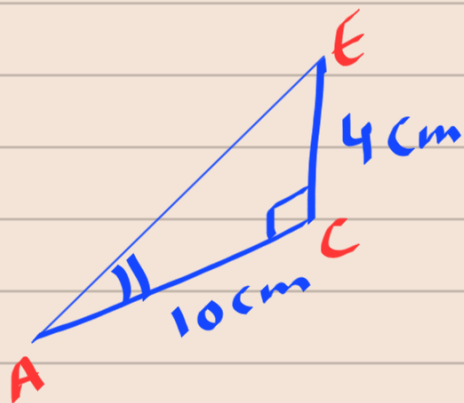
$$EC = 4 \text{ cm}$$

Find the size of the angle between

↓
AE and the plane ABCD??



$$AC = \sqrt{6^2 + 8^2} = 10 \text{ cm}$$



$$\tan A = \frac{4}{10}$$

$$\angle EAC = \tan^{-1}\left(\frac{4}{10}\right) = 21.8^\circ$$